Tube Rupture Relief Load: Custom HEM Model vs. GPREH Textbook Answer

GPREH Appendix D.1.2 – vertical thermosyphon reboiler, propane tube side, bottom-tubesheet rupture 6 in from the tube end. Comparison of the from-scratch Homogeneous Equilibrium Model (HEM) pipe-flow solver (built as a stand-in for DIERS TPHEM, driven by *thermo* PR-EOS property lookups) against the textbook's published solution.

1. Part 1 Result — Relief Load Through the Break

Both approaches split the ruptured tube into a short liquid leg (0.5 ft, upward, fed from 9 ft of column liquid head) and a long two-phase leg (11.5 ft, downward, 15% vapor quality, gravity-assisted). The textbook's per-leg split is inferred algebraically from the Part 2 flow-weighted-enthalpy equation (only the Part 2 answer page was available for this comparison, not Part 1's own worked solution) and should be treated as approximate.

Leg	This model	Textbook (inferred)	Difference
Short leg (liquid, 0.5 ft, up)	33,183 lb/hr	29,952 lb/hr*	+10.8%
Long leg (2-phase, 11.5 ft, down)	18,304 lb/hr	20,773 lb/hr*	-11.9%
Total relief load	**51,487 lb/hr**	**50,725 lb/hr**	**+1.5%**

*Inferred from the Part 2 equation's flow-weighting terms, not read directly from a Part 1 answer.

The total relief load agrees to within 1.5% – a reasonable validation of the custom HEM solver given it uses a completely different property source (Peng-Robinson EOS via *thermo*) and numerical method (shooting-method mass-flux bisection) than the textbook's TPHEM/chart-based approach. The individual legs, however, disagree by about 11% in opposite directions, and the errors happen to cancel in the total. This is the more interesting finding: the two methods apportion the same total flow differently between the choked liquid leg and the backpressure-limited two-phase leg, most likely reflecting differences in the friction correlation, entrance-loss treatment, or PR-EOS vs. chart-based density/enthalpy data along each leg's expansion path.

2. Part 2 — Vapor Fraction at the Shell Relieving Pressure

The textbook's Part 2 vapor fraction and this model's per-leg exit qualities are answering **different questions** and should not be read as disagreeing with each other:

- **Textbook (34.81% vapor):** mixes the two legs' flows into one enthalpy (45.57 Btu/lbm at 300 psig, using h_f=31.9 / h_g=141.6 Btu/lbm), then performs a static isenthalpic flash of that combined stream down to 179.7 psia (165 psig) – i.e. the fully velocity-relaxed thermodynamic state after the jet has fully expanded and dissipated its kinetic energy back to enthalpy inside the shell.
- **This model (mass-flow-weighted ~22.2% vapor: 14.3% short leg / 36.4% long leg):** these are the local flowing qualities *at the pipe exit*, computed from the HEM energy balance h(P,v) = h■ – (Gv)²/2. A meaningful share of the enthalpy drop at that point is still kinetic energy of the high-velocity jet (~11,000 kg/(m²·s) and ~6,000 kg/(m²·s) mass flux for the short and long legs respectively), so less of it has converted to vapor yet.

Once the jet exits into the shell and its velocity dissipates, that kinetic energy converts back to enthalpy and drives further flashing – consistent with the textbook's higher, fully-relaxed vapor fraction (34.81%) exceeding this model's velocity-retained pipe-exit qualities (14–36%, ~22% flow-weighted average). The two numbers are compatible, not contradictory: they describe the fluid at two different points along its expansion (in-pipe vs. post-jet-dissipation in the shell).

3. Conclusion

The custom PR-EOS-driven HEM solver reproduces the textbook's total relief load (51,487 vs. 50,725 lb/hr, +1.5%) closely enough to serve as a working substitute for TPHEM on this workstation. The per-leg flow split and the vapor-fraction figures differ for identifiable, explainable reasons (friction/EOS methodology and kinetic-energy accounting, respectively) rather than indicating a modeling error.